

Unit 1

Semiconductor Diodes and their Applications:

What is Electronics ?

- Electronics is the branch of science and engineering which deals with the study of flow of electrons through vacuum, gas, or semiconductor.
- An electronic circuit is a closed path formed by interconnection of electronic components through which electric current flows to perform a specific function

Types of Electronics:

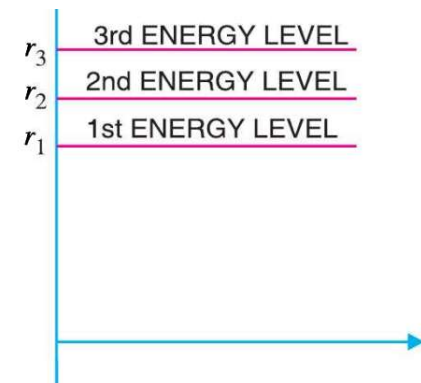
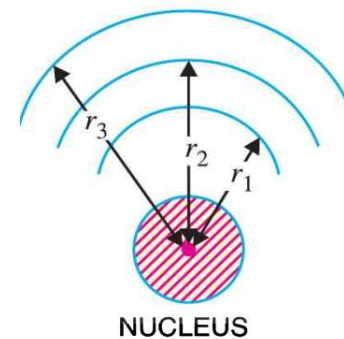
- Analog Electronics
- Digital Electronics
- Power Electronics
- Microelectronics
- Optoelectronics
- Communication Electronics

Atomic Structure

- An atom is the smallest particle of an element that retains the chemical properties of that element.
- Atomic structure refers to the structure of atom comprising of a nucleus (center) in which the protons (positively charged) and neutrons (neutral) are present.
- The negatively charged particles called **electrons revolve around the center of the nucleus**.
- The electrons in each permitted orbit have a certain fixed amount of energy. The larger the orbit (i.e. larger radius), the greater is the energy of electrons.
- If an electron is given additional energy (e.g. heat, light etc.), it is lifted to the higher orbit. The atom is said to be in a state of excitation.

Atomic weight = no of protons + no of neutrons.

Atomic number = no of protons = no of electrons in atom



Semiconductor

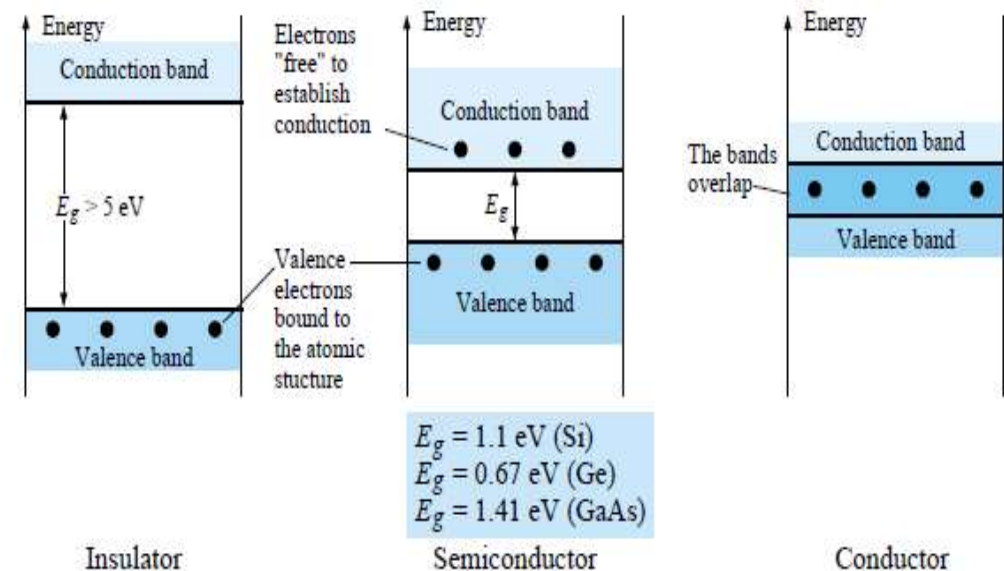
- In terms of their electrical properties, materials can be classified into three groups: conductors, semiconductors, and insulators.
- A **semiconductor** is a material whose **electrical conductivity lies between that of a conductor and an insulator** and whose conductivity can be **controlled** by adding impurities (doping) or applying external fields.
- The behavior of electrons in solids is governed by their arrangement in energy bands.

Conduction Band: The energy band where electrons are free to move and contribute to electrical conductivity.

Valence Band: The energy band where electrons are bound to atoms and do not contribute to conductivity.

Fermi Level: The highest energy level that an electron can occupy at the absolute zero temperature is known as the Fermi Level.

The Fermi level lies between the valence band and conduction band

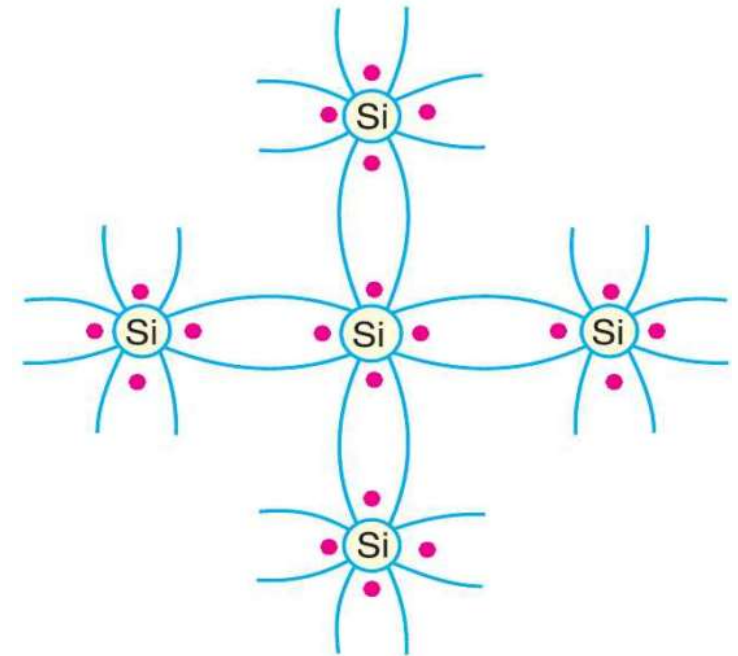
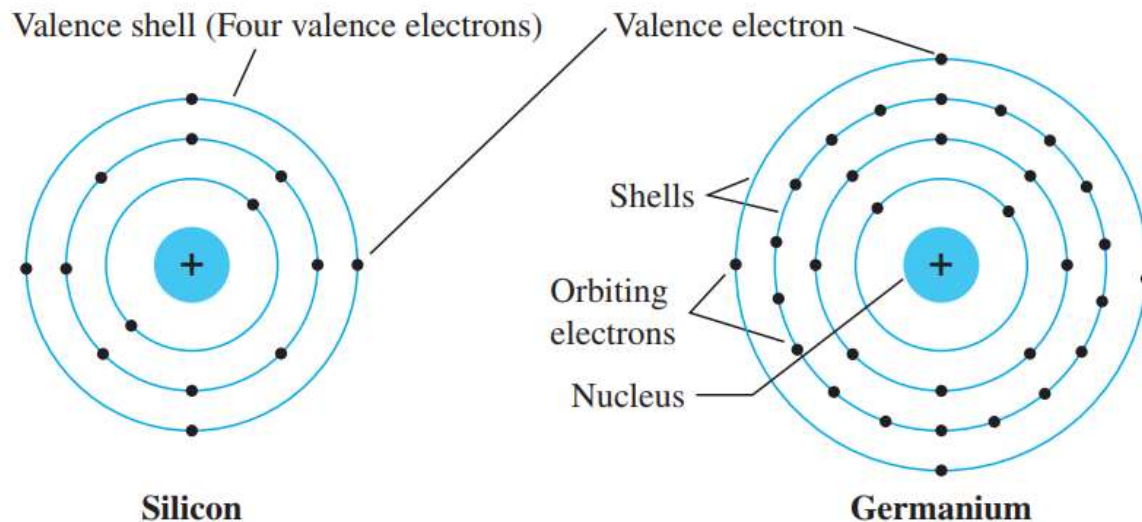


Comparison Table of Materials

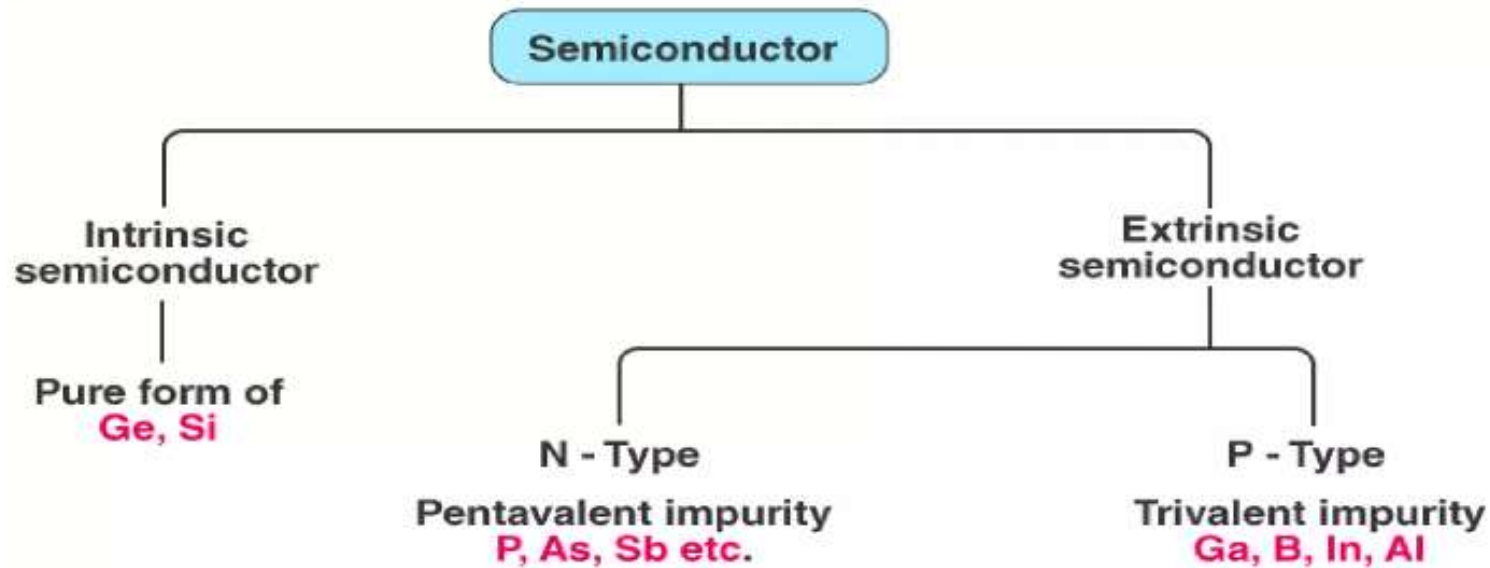
Parameter	Conductor	Semiconductor	Insulator
Forbidden Energy Gap	Not Exist	Small (1eV)	Large (>5eV)
Resistivity	Low	Moderate	High
Flow of current	Due to movement of free electrons.	Due to movement of electrons and holes.	Almost negligible but only due to free electrons.
Charge carriers in conduction band	Completely filled	Partially filled	Completely vacant
Charge carriers in valence band	Almost vacant	Partially filled	Completely filled
Example	Copper, Aluminium, graphite etc.	Silicon, Germanium, arsenic etc.	Paper, rubber, glass, plastic etc.
Applications	Conducting wires, Transformers, in electrical cords etc.	Diodes, transistors, opto couplers etc.	Sports equipment, home appliances etc.

Semiconductor

- In general, semiconductor materials fall into one of two classes: *single-crystal* and *compound*.
- The three semiconductors used most frequently in the construction of electronic devices are ***Ge, Si, and GaAs***.
- In a pure silicon or germanium crystal, the four valence electrons of one atom form a bonding arrangement with four adjoining atoms ***is called covalent bonding***.



Types of Semiconductors



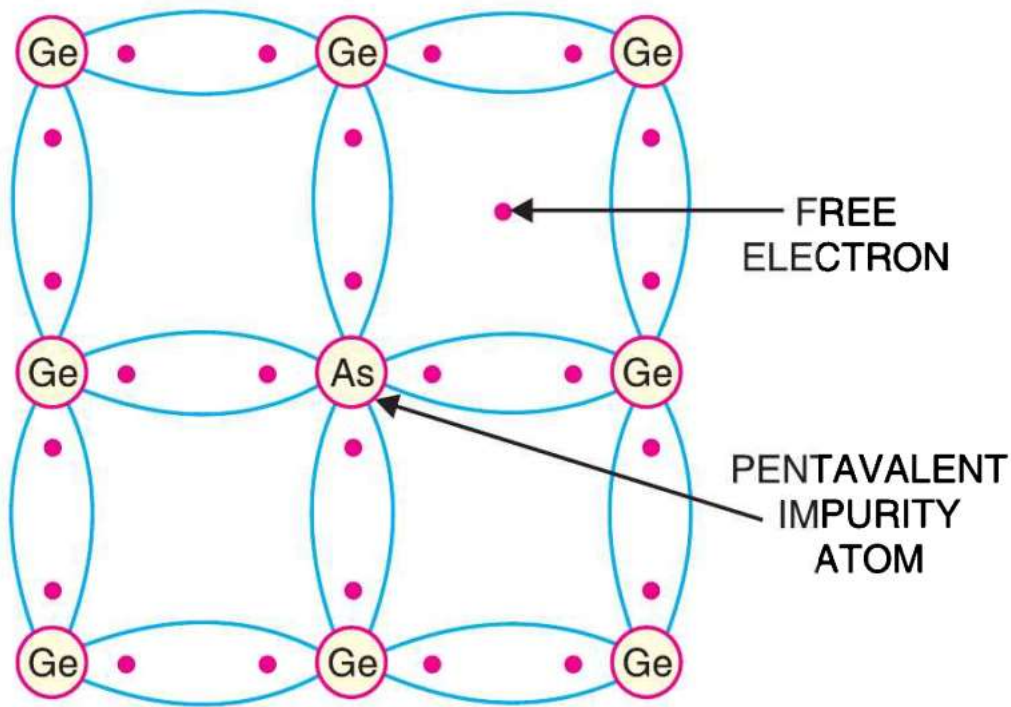
Intrinsic semiconductors

- Intrinsic semiconductors are pure form semiconductor materials without any significant impurities. Examples Silicon (Si) and germanium (Ge).
- In an intrinsic semiconductor, the number of electrons in the conduction band (where they can move freely) equals the number of holes (empty states in the valence band where an electron could exist).

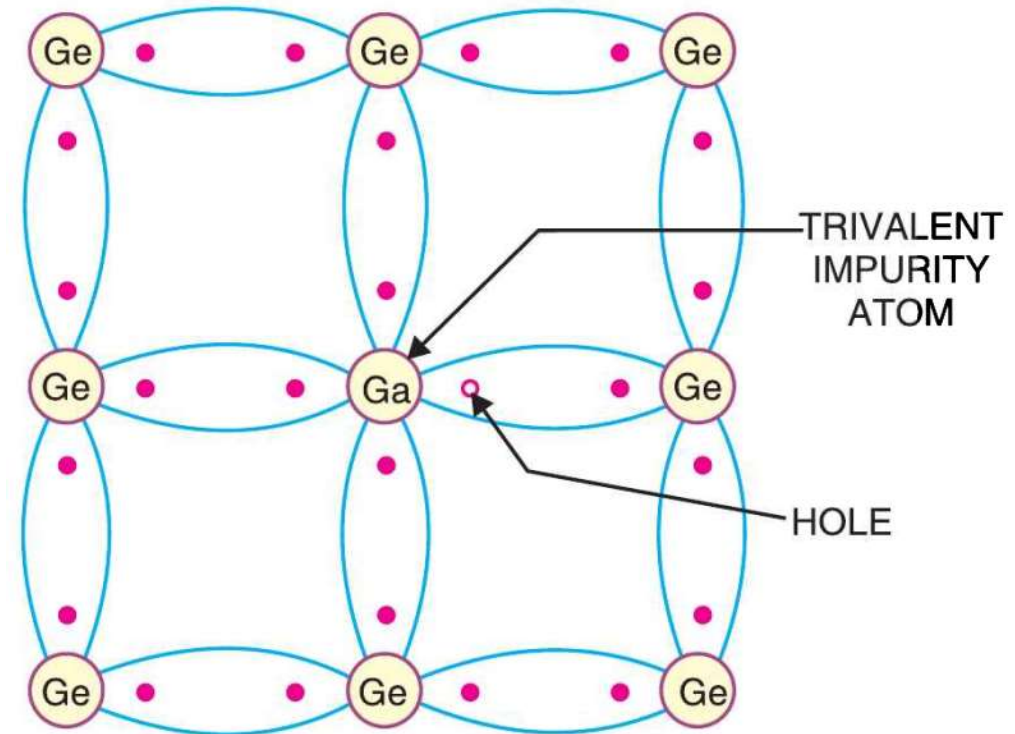
Extrinsic Semiconductors

- Extrinsic semiconductors are created by adding specific impurities, called dopants, to an intrinsic semiconductor to enhance its conductivity.
- Types of Extrinsic Semiconductors:
 - ✓ **n-type:** An *n*-type material is created by adding impurity elements that have *five* valence electrons (***pentavalent***), such as *antimony*, *arsenic*, and *phosphorus*. *The diffused impurities with five valence electrons are called donor atoms.*
 - ✓ **p-type:** An *p*-type material is created by adding impurity elements that have *three* valence electrons (***trivalent***), such as *boron*, *gallium*, and *indium*. *The diffused impurities with three valence electrons are called acceptor atoms.*
- The controlled addition of dopants allows precise control over the electrical properties of semiconductors, making them essential in devices like diodes, transistors, and integrated circuits.

Extrinsic Semiconductors



Arsenic impurity in n-type material



Gallium impurity in p-type material

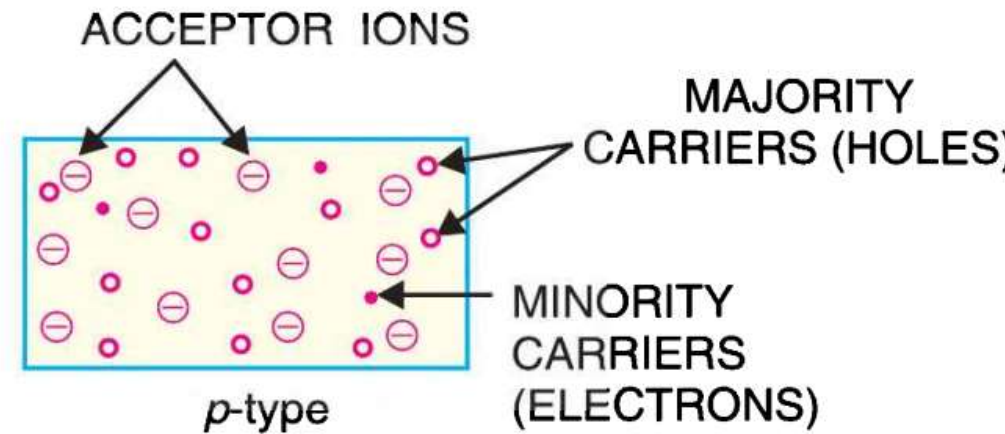
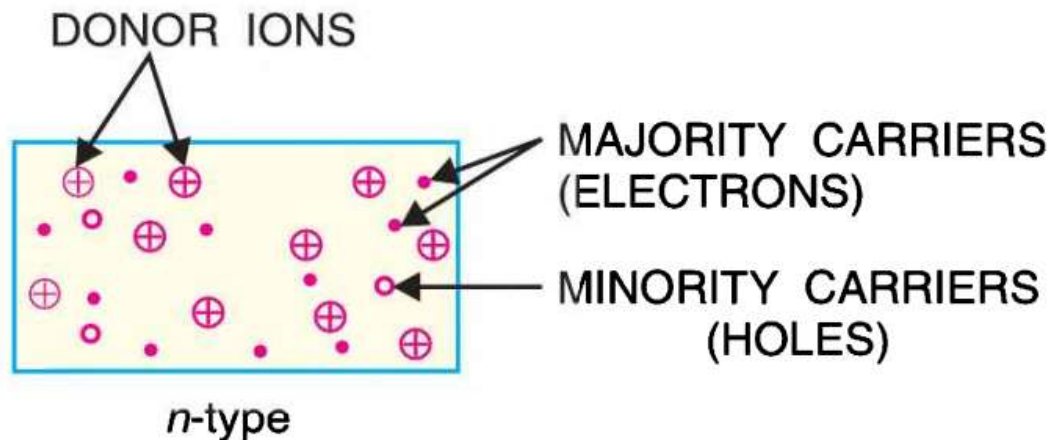
Intrinsic Semiconductors Vs Extrinsic Semiconductors

Property	Intrinsic Semiconductor	Extrinsic Semiconductor (n-type & p-type)
Purity	Pure (no impurities)	Doped with impurities
Conductivity	Low at room temperature	Higher due to the presence of free charge carriers
Charge Carriers	Equal number of electrons and holes	Majority and minority carriers (electrons or holes)
Effect of Temperature	Conductivity increases with temperature	Conductivity depends on doping and temperature
Types	Not applicable	n-type (electron majority) and p-type (hole majority)
Energy Band Structure	No additional energy levels	Donor or acceptor energy levels present

Majority and Minority Carriers

*In an **n-type** material the **electron** is called the **majority** carrier and the **hole** the **minority** carrier.*

*In a **p-type** material the **hole** is the **majority** carrier and the **electron** is the **minority** carrier*



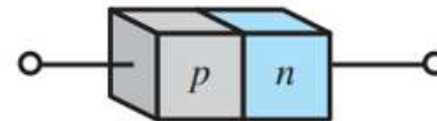
PN Junction Diode

- The **PN junction** is the fundamental building block of semiconductor devices such as diodes, transistors, and integrated circuits.
- The PN junction diode is created by simply joining an n -type and a p -type material together.
- **A diode has three operating conditions:**

No bias

Forward bias

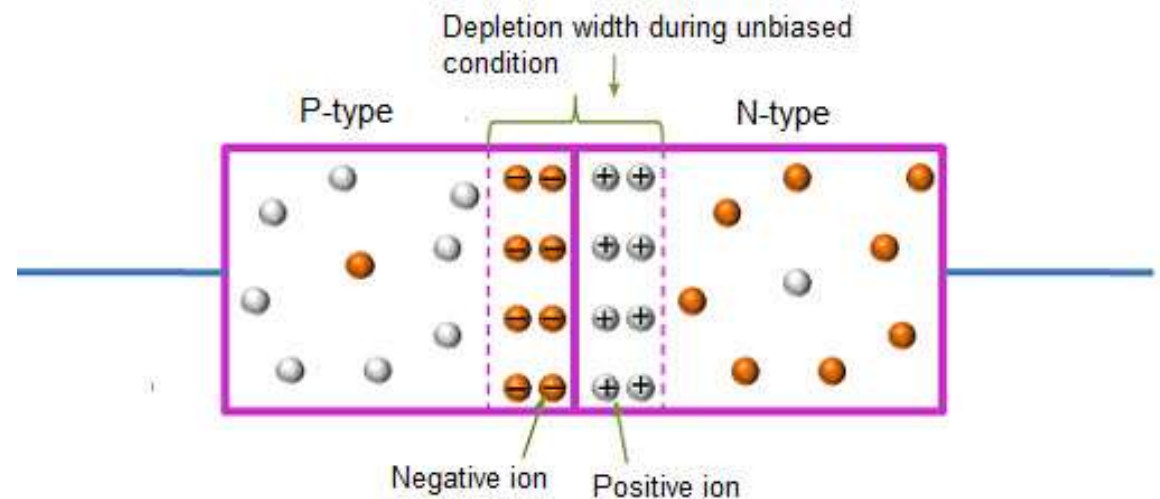
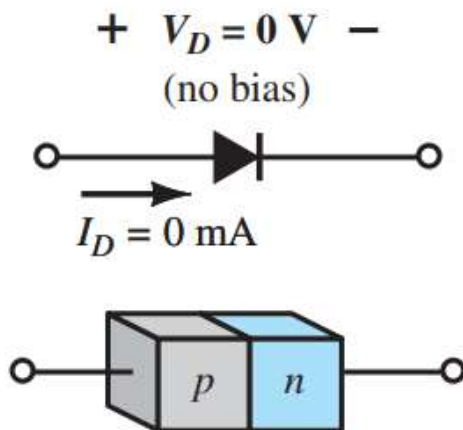
Reverse bias



- The term *bias* refers to the application of an external voltage across the two terminals of the device to extract a response.

PN Junction :No Bias

- Electrons from n-region diffuse into p-region. Holes from p-region diffuse into n-region. Movement continues until balance is established.
- Electrons and holes recombine near junction, creating a region with no free charge carriers. Only immobile ions remain (negative ions on p-side, positive ions on n-side).
- An electric field develops across the depletion region. The flow of electrons and holes due to diffusion is balanced by the force of this electric field.
- In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero.



Characteristics of the Depletion Region

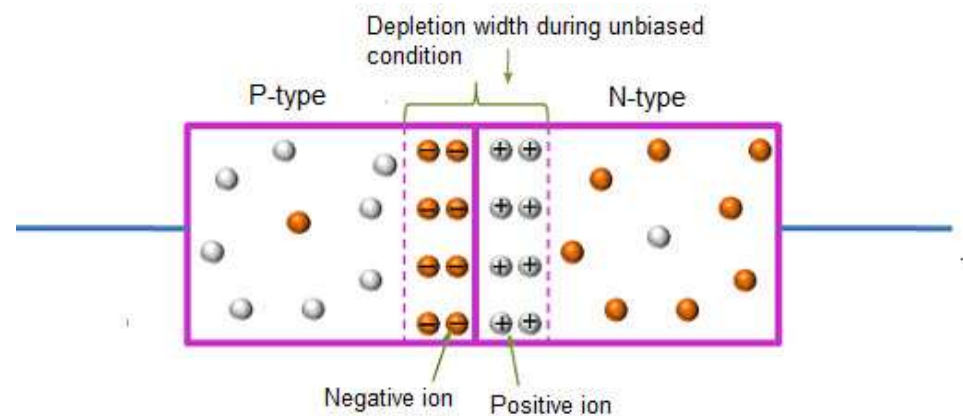
- **Width:** Depends on doping concentration. Heavily doped regions have narrower depletion regions.
- **Charge Distribution:** p-region contains negatively charged acceptor ions; n-region contains positively charged donor ions.
- **Electric Field:** Points from positively charged n-region to negatively charged p-region.

Barrier Potential

The electric field creates a potential barrier preventing further diffusion of charge carriers.

Built-in potential (V_b) is typically:

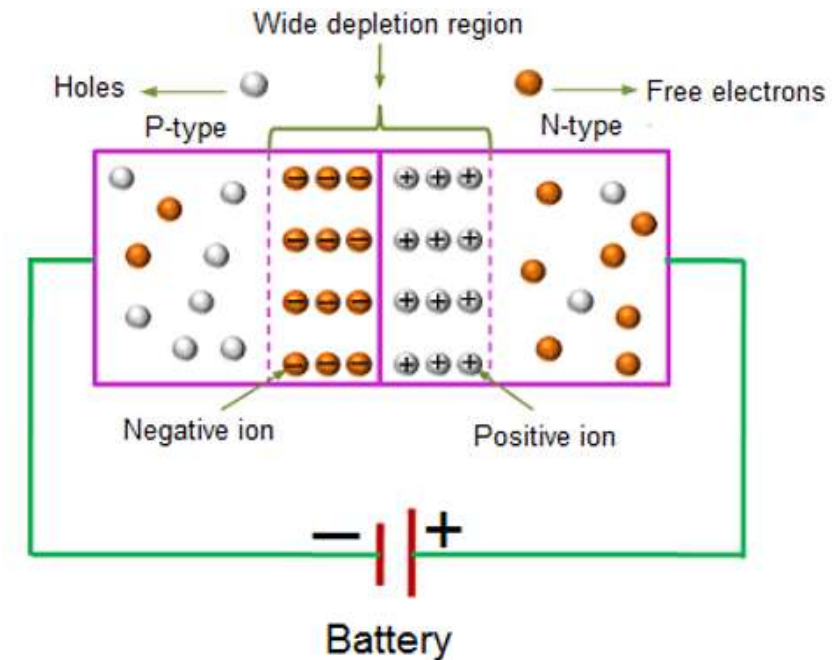
- 0.7V for silicon
- 0.3V for germanium



Biasing of PN Junction: Reverse Bias

When positive voltage is applied to n-region and negative to p-region:

- Potential barrier is **increased**
- Depletion region widens
- Charge carriers cannot cross the junction
- Very little current flows (minority charge carrier)
- If reverse voltage exceeds critical value, breakdown can occur through Zener or avalanche breakdown

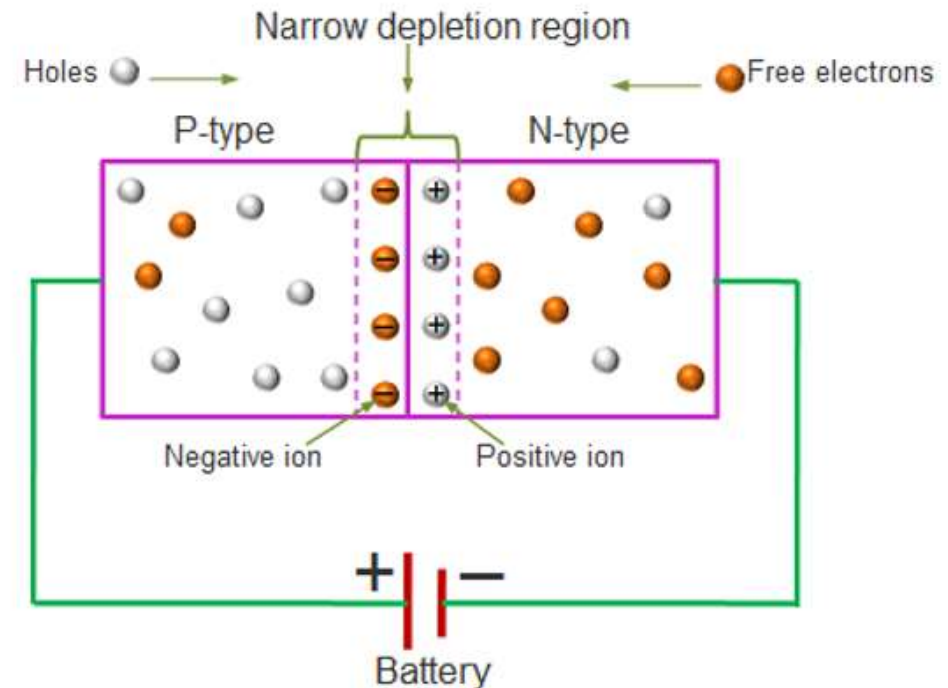


- Reverse saturation current (I_s) or leakage current is caused by thermally generated minority carriers swept across the junction by the built-in electric field.
- It is a very small, almost constant with reverse voltage and strongly increases with temperature.

Biasing of PN Junction: Forward Bias

When positive voltage is applied to p-region and negative to n-region

- Potential barrier is **reduced**
- Charge carriers can cross the junction
- Electrons move toward p-region, holes toward n-region
- Results in **forward current** flow
- In silicon diodes, forward voltage must be $\sim 0.7\text{V}$ for significant current.



Biasing of PN Junction: Forward Bias

General characteristics of a semiconductor diode : Shockley's equation

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (\text{A})$$

I_s is the reverse saturation current

V_D is the applied forward-bias voltage across the diode

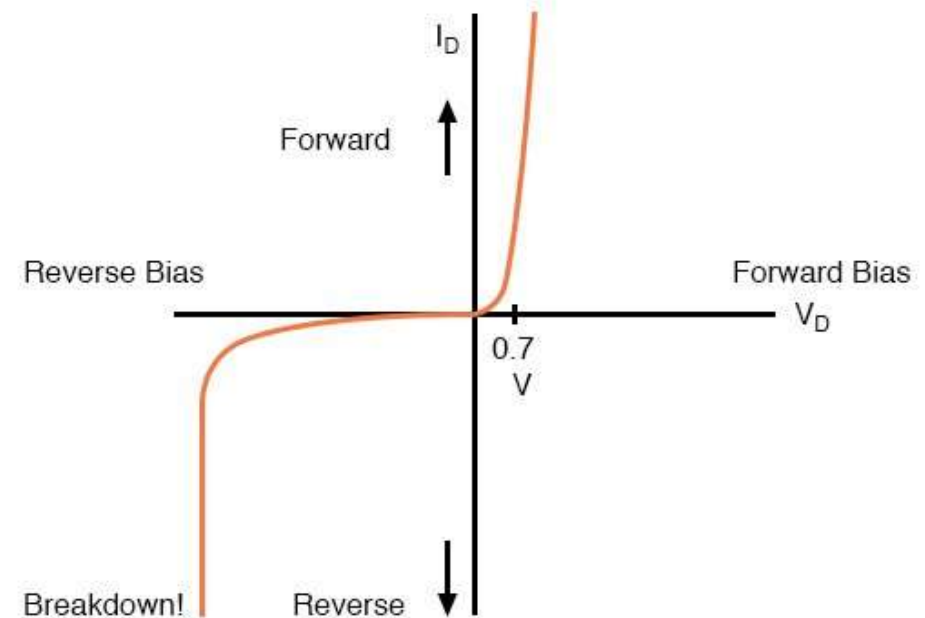
n is the ideality factor, and $n=1$ for Germanium and $n=2$ for Silicon

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

k is Boltzmann's constant = 1.38×10^{-23} J/K

T_K is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$

q is the magnitude of electronic charge = 1.6×10^{-19} C



Current-Voltage (I-V) Characteristics

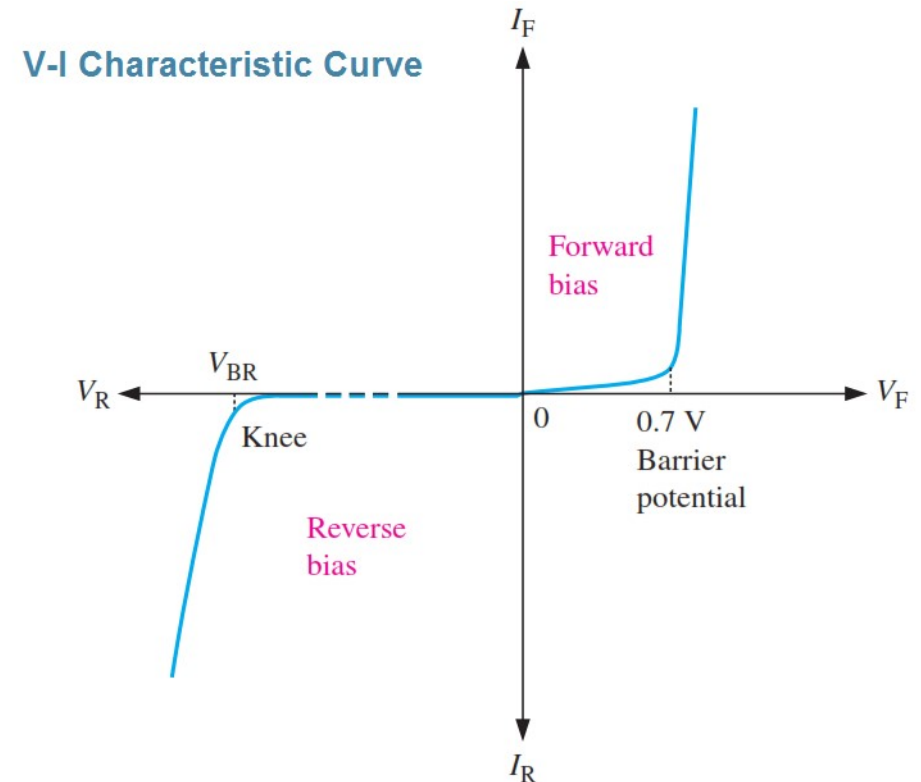
Forward Bias

- When voltage exceeds threshold (0.7V for silicon), current increases exponentially
- Diode conducts current easily

Reverse Bias

- Only very small reverse saturation current flows
- If reverse voltage exceeds breakdown voltage, diode conducts large currents (may damage regular diodes)

Knee Voltage: It is the minimum voltage after which the forward current starts increasing exponentially is called knee voltage



Thermal Voltage Calculation at Room Temperature

Question : At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage V_T .

Answer: The thermal voltage V_T can be calculated as

$$V_T = \frac{kT_K}{q}$$

(V)

k is Boltzmann's constant = 1.38×10^{-23} J/K

T_K is the absolute temperature in kelvins = 273 + the temperature in °C

q is the magnitude of electronic charge = 1.6×10^{-19} C

$$\begin{aligned} T &= 273 + ^\circ\text{C} = 273 + 27 = 300 \text{ K} \\ V_T &= \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \\ &= 25.875 \text{ mV} \cong 26 \text{ mV} \end{aligned}$$

Applications of PN Junction



Diodes

Allow current to flow in only one direction



Rectifiers

Convert AC to DC in power supply circuits



LEDs

When forward biased, energy is released as light



Solar Cells

Convert light energy into electrical energy using photovoltaic effect



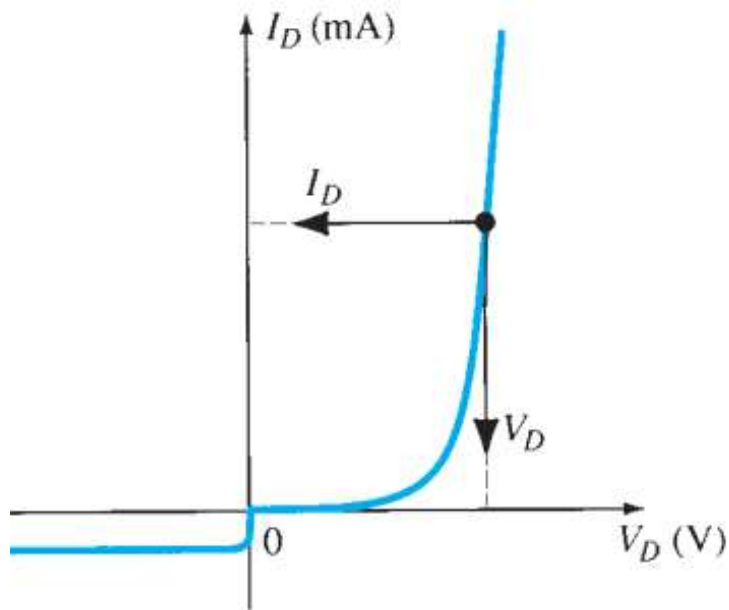
Transistors

Form the basis of BJTs and FETs

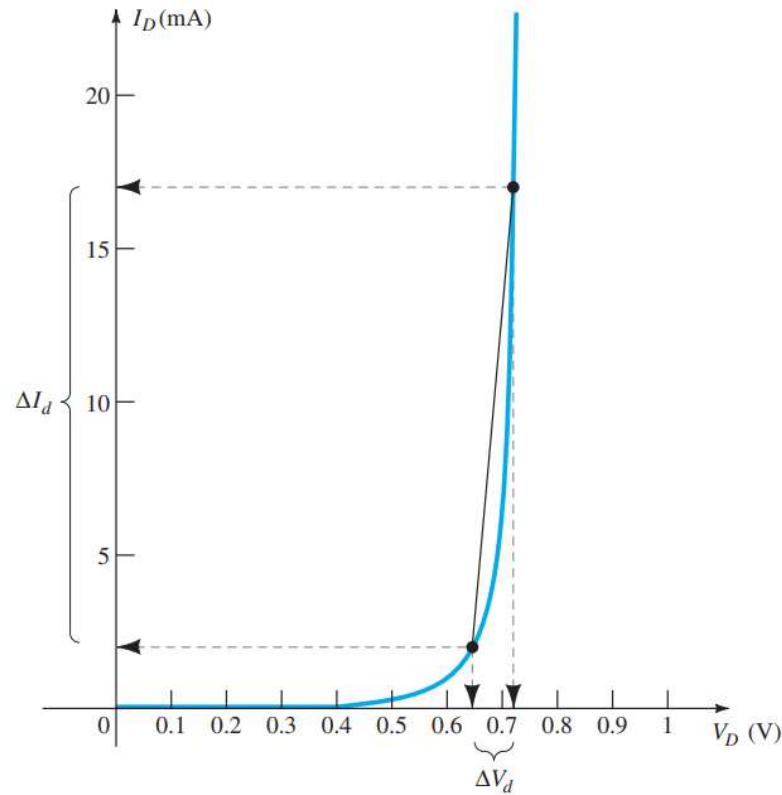
Diode Resistance

Static Resistance / DC Resistance: Defined as a point on the characteristics

Dynamic Resistance / AC resistance : AC resistance can be calculated using the current and voltage values for two points on the diode characteristics



$$R_D = \frac{V_D}{I_D}$$



$$r_d = \frac{\Delta V_d}{\Delta I_d}$$

$$\frac{dV_D}{dI_D} = r_d = \frac{nV_T}{I_D}$$

$$n = 1 \text{ and } V_T \cong 26 \text{ mV}$$

$$r_d = \frac{26 \text{ mV}}{I_D}$$

Temperature Effects

As temperature increases it adds energy to the diode.

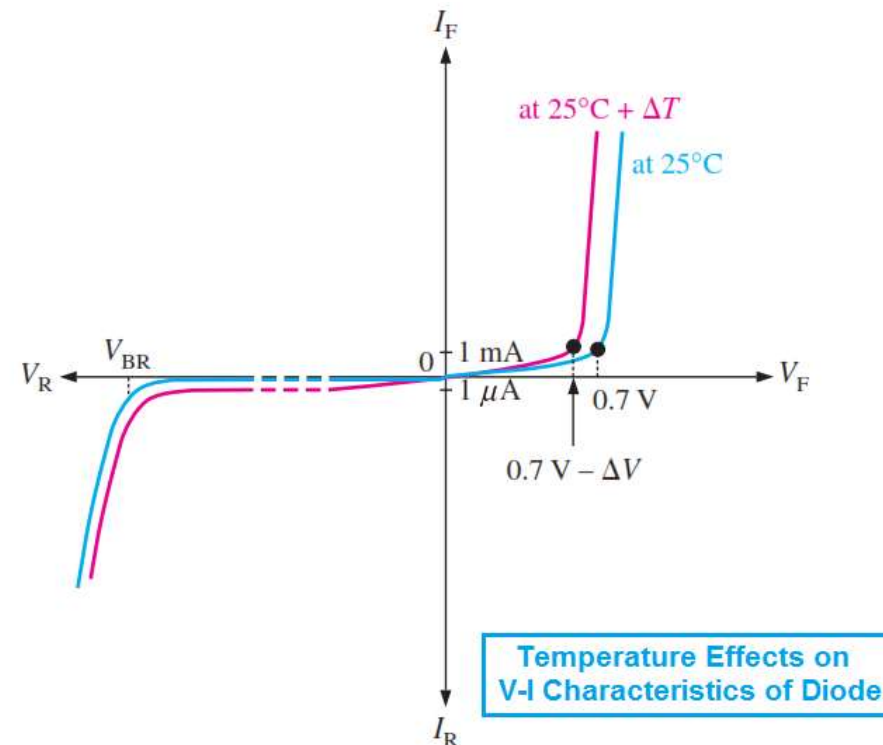
- It reduces the required forward bias voltage for forward bias conduction.
- It increases the amount of reverse current in the reverse bias condition.

Germanium diodes are more sensitive to temperature variations than silicon or gallium arsenide diodes.

In the forward bias region, the characteristics of Silicon diode shift to the left (lower voltage) at a rate of $2.5\text{ mV} / ^\circ\text{C}$ increase in temperature.

Reverse saturation current **approximately doubles for every 10°C rise** in temperature.

$$I_S \approx I_{S0} 2^{\frac{T-T_0}{10}}$$



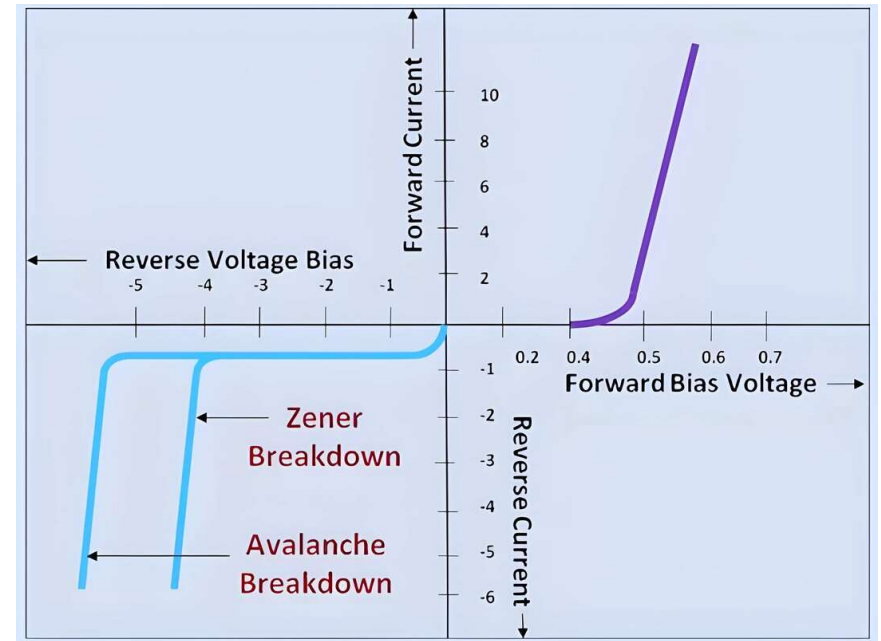
Breakdown Mechanism

Avalanche break down:-

- If both p-side and n-side of the diode are **lightly doped**, depletion region at the **junction widens**.
- Application of a very large electric field at the junction may rupture covalent bonding which leads to the generation of a large number of charge carriers resulting in avalanche multiplication.

Zener break down:-

- If both p-side and n-side of the diode are **heavily doped**, depletion region at the **junction reduces**.
- Application of even a **small voltage** at the junction **ruptures covalent** bonding and generates large number of charge carriers.
- Such sudden increase in the number of charge carriers results in Zener mechanism.



Zener breakdown needs less voltage at higher temperature, while avalanche breakdown needs more voltage at higher temperature.

Zener breakdown → Negative temperature coefficient
Avalanche breakdown → Positive temperature coefficient

Diode Capacitance

Depending upon the biasing condition, two types of capacitive effects exist in the diodes namely **Junction Capacitance and Diffusion Capacitance**

Junction Capacitance (or Depletion Capacitance)

- It is the most significant capacitance and varies with the reverse bias voltage applied.
- Junction capacitance is associated with the depletion region of a diode.
- As the reverse bias voltage increases, the width of the depletion region increases, leading to a decrease in junction capacitance.
- The junction capacitance is given by

$$C_j = \frac{\epsilon A}{W}$$

ϵ = the permittivity of the dielectric

A=Area of the junction

W= Width of the depletion

Diffusion Capacitance:

- Diffusion capacitance appears in a **forward-biased PN junction** due to **storage of charge** in the diffusion region of the diode.
- In forward biased condition, the width of the depletion region decreases. Thus, as the applied voltage increases, concentration of injected charged particles increases.
- This rate of change of the injected charge with applied voltage is defined as a capacitance called diffusion capacitance.
- Diffusion capacitance increases with an increase in forward current. It is significant only when the diode is forward-biased.
- It is typically less than junction capacitance in most applications.

The diffusion capacitance is given by

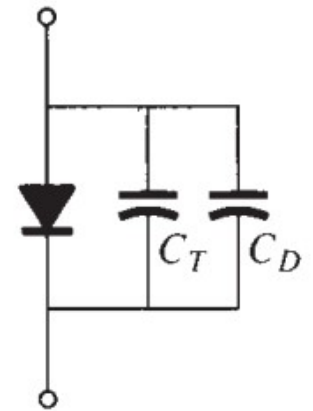
$$C_d = \frac{\tau I_D}{V_T}$$

where

τ = Carrier Life time

I_D = Diode Current

V_T = Thermal Voltage



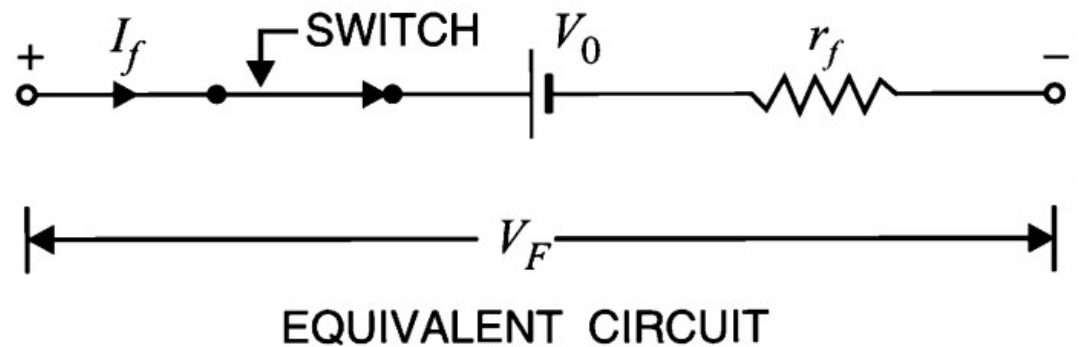
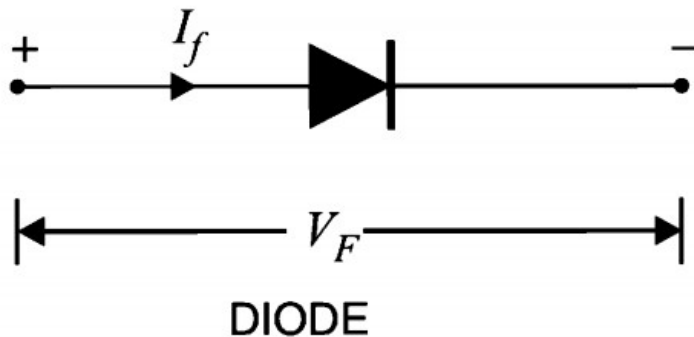
effect of the transition or diffusion capacitance on the diode

Equivalent Model of Diode

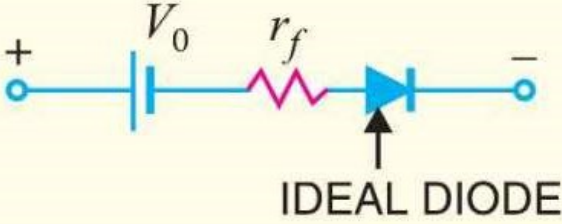
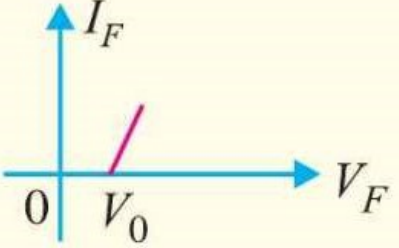
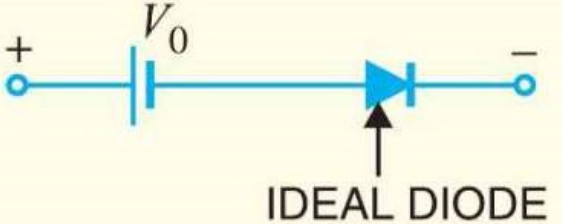
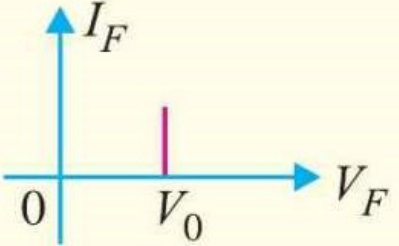
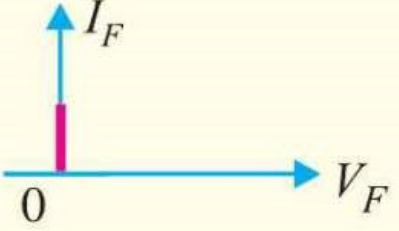
The forward voltage V_F applied across the actual diode has to overcome

- (a) potential barrier V_0
- (b) internal drop $I_f r_f$

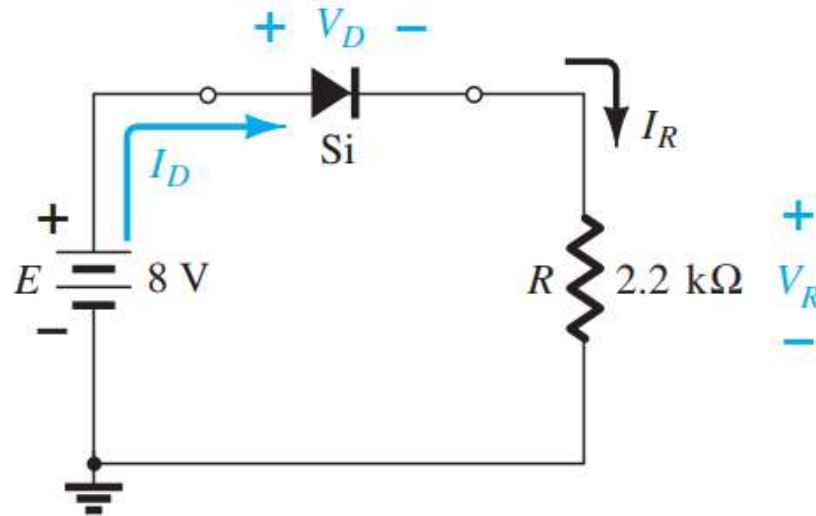
$$V_F = V_0 + I_f r_f$$



Equivalent Model of Diode

S.No.	Type	Model	Characteristic
1.	Approximate model		
2.	Simplified model		
3.	Ideal Model		

For the series diode configuration of Fig, determine V_D , V_R , and I_D .



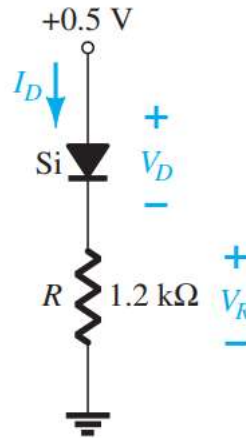
Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = \mathbf{0.7\text{ V}}$$

$$V_R = E - V_D = 8\text{ V} - 0.7\text{ V} = \mathbf{7.3\text{ V}}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong \mathbf{3.32\text{ mA}}$$

For the series diode configuration of Fig, determine V_D , V_R , and I_D .



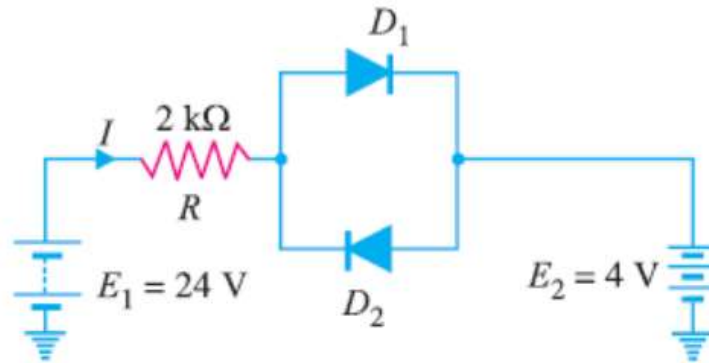
Although the “pressure” establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode “on.”. The resulting voltage and current levels are therefore the following

$$I_D = \mathbf{0\text{ A}}$$

$$V_R = I_R R = I_D R = (0\text{ A}) 1.2\text{ k}\Omega = \mathbf{0\text{ V}}$$

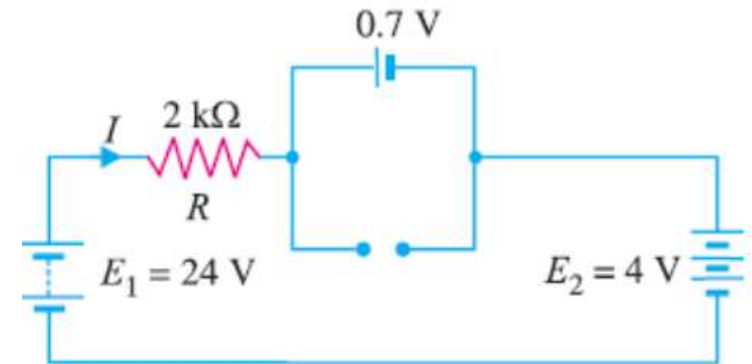
$$V_D = E = \mathbf{0.5\text{ V}}$$

Determine the current I in the circuit shown in Fig. Assume the diodes to be of silicon and forward resistance of diodes to be zero.

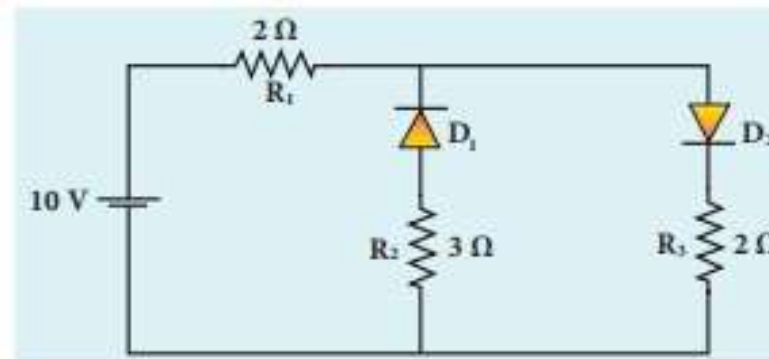


From the Fig, it is clear that that diode D_1 is forward biased and diode D_2 is reverse biased.

$$I = \frac{E_1 - E_2 - V_0}{R} = \frac{24 - 4 - 0.7}{2\text{ k}\Omega} = \frac{19.3\text{ V}}{2\text{ k}\Omega} = 9.65\text{ mA}$$



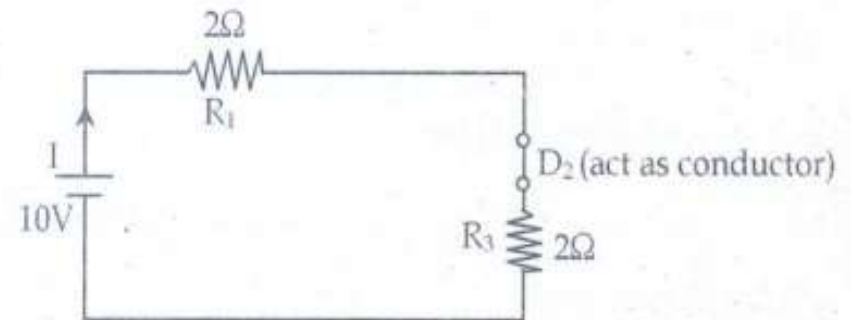
Q. The given circuit has two ideal diodes connected as shown in figure below. Calculate the current flowing through the resistance R₁



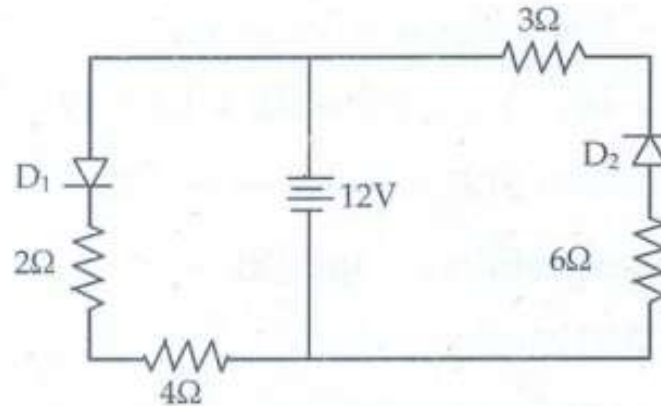
Ans : Barrier potential for ideal diode is zero. The diode D₁ is reverse biased, so it will block the current and diode D₂ is forward biased, so it will pass the current.

$$\text{Effective resistance } R_{\text{eff}} = R_1 + R_3 = 4\Omega$$

$$\text{Current through } R_1 = V/R_{\text{eff}} = 10/4 = 2.5\text{A}$$



Determine the current flowing through 3Ω and 4Ω resistors of the circuit given below. Assume that diodes D_1 and D_2 are ideal diodes.



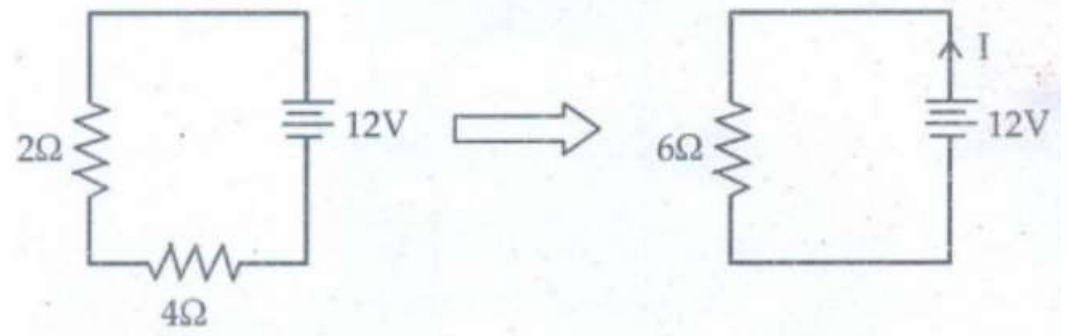
Solution : The diode D_2 is in reverse biased. So does not conduct current.

$\therefore$ current through 3Ω is $= 0$

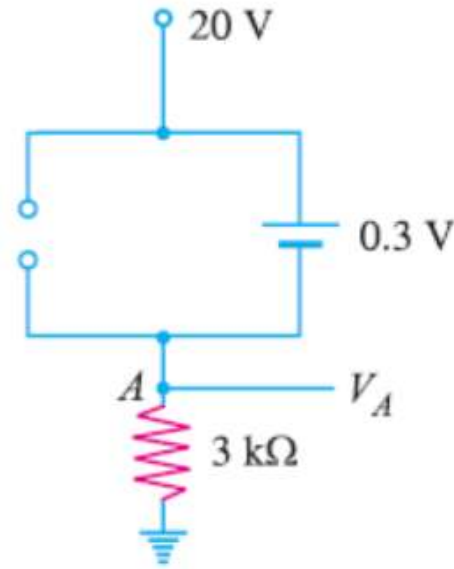
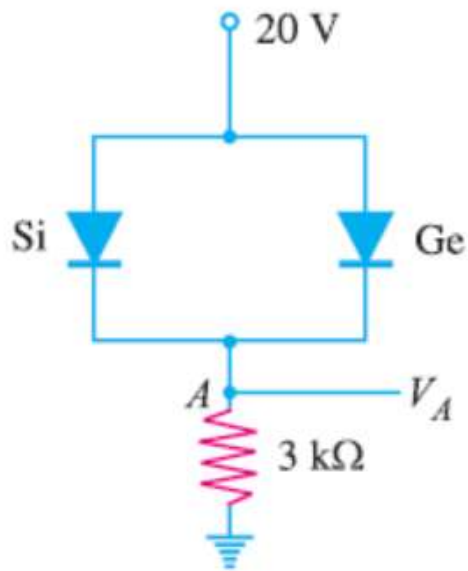
The diode D_1 is in forward biased, and it is an ideal diode.

The current through 4Ω is

$$\therefore I = V/R = 12/6 = 2A$$



Find the voltage V_A in the circuit shown in Fig. Use simplified model.

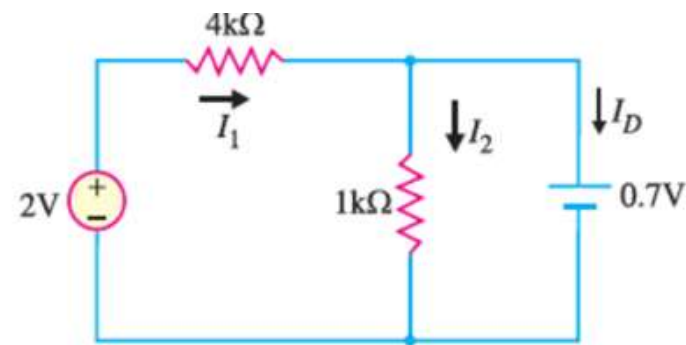
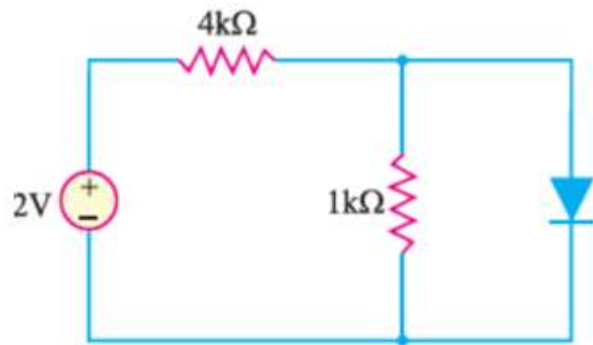


When voltage is applied, germanium diode ($V_0 = 0.3 \text{ V}$) will turn on first and a level of 0.3V is maintained across the parallel circuit.

The silicon diode never gets the opportunity to have 0.7 V across it and, therefore, remains in open-circuit state

$$V_A = 20 - 0.3 = 19.7 \text{ V}$$

Determine the state of diode for the circuit shown in Fig. and find I_D and V_D . Assume simplified model for the diode.



Let us assume that the diode is ON. Therefore, we can replace the diode with a 0.7V battery as shown in Fig.

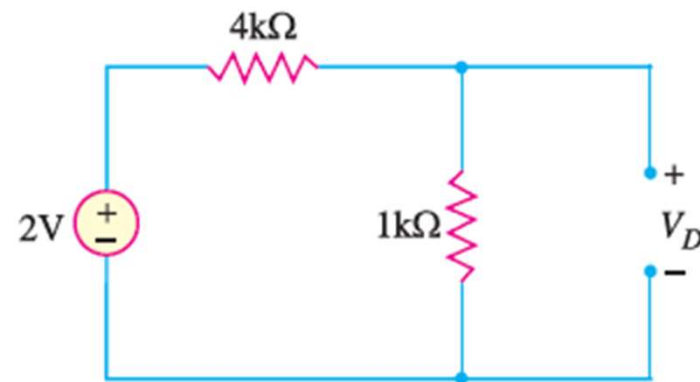
$$I_1 = \frac{(2 - 0.7) \text{ V}}{4 \text{ k}\Omega} = \frac{1.3 \text{ V}}{4 \text{ k}\Omega} = 0.325 \text{ mA}$$

$$I_2 = \frac{0.7 \text{ V}}{1 \text{ k}\Omega} = 0.7 \text{ mA}$$

$$I_D = I_1 - I_2 = 0.325 - 0.7 = -0.375 \text{ mA}$$

Since the diode current is negative, the diode must be OFF and the true value of diode current is $I_D = 0 \text{ mA}$. Hence our initial assumption was wrong.

In order to analyse the circuit properly, we should replace the with an open circuit as shown in Fig.



The voltage V_D across the diode is

$$V_D = \frac{2 \text{ V}}{1 \text{ k}\Omega + 4 \text{ k}\Omega} \times 1 \text{ k}\Omega = 0.4 \text{ V}$$

We know that 0.7V is required to turn ON the diode. Since V_D is only 0.4V, the answer confirms that the diode is OFF.

A germanium diode has a reverse saturation current of $0.19\mu\text{A}$. Find the current in the diode when it is forward biased with 0.3V at 27°C .

$$V_T = \frac{kT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23} T}{1.6 \times 10^{-19}}$$

$$V_T = 0.8625 \times 10^{-4} T$$

$$V_T = 8.625 \times 10^{-5} T \text{ volts}$$

$$8.625 \times 10^{-5} \approx \frac{1}{11600}$$

$$V_T = \frac{T}{11600}$$

Given:

$I_o = 0.19\mu\text{A}$, $V_D = 0.3\text{V}$, $T = 27^\circ\text{C}$, $\eta = 1$ for Germanium Diode, and $I_D = ?$

$$I_D = I_o \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow I_D = 0.19 \times 10^{-6} \times \left(e^{0.3\text{V} / 1 \times 25.862 \times 10^{-3}} - 1 \right)$$

$$\Rightarrow I_D = \mathbf{20.72 \text{ mA}}$$

$$\begin{aligned} V_T &= \frac{T}{11,600} \\ &= \frac{273^\circ + 27^\circ}{11,600} \\ &= 25.862\text{mV} \end{aligned}$$

The forward current in a Silicon Diode is 15mA at 27°C. If reverse saturation current is 0.24nA, what is the forward bias voltage?

$$I_D = 15mA, I_o = 0.24nA, T = 27^\circ C, \eta = 2 \text{ for Si Diode, and } V_D = ?$$

$$I_D = I_o \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow V_D = \eta V_T \ln \left(\frac{I_D}{I_o} + 1 \right)$$

$$\Rightarrow V_D = 2 \times 25.862mV \times \ln \left(\frac{15m}{0.24n} + 1 \right)$$

$$V_D = \mathbf{0.928V}$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 27^\circ}{11,600}$$

$$= 25.862mV$$

A germanium diode carries a current of 10mA when it is forward biased with 0.2V at 27°C. (a) Find reverse saturation current. (b) Find the bias voltage required to get a current of 100mA.

Given, $I_D = 10mA$, $V_D = 0.2V$, $t = 27^\circ C$, $\eta = 1$ for Ge Diode

(a) $I_o = ?$

$$I_o = \frac{I_D}{\left(e^{V_D / \eta V_T} - 1 \right)}$$

$$I_o = \frac{10m}{\left(e^{0.2 / 1 \times 25.862m} - 1 \right)} \Rightarrow I_o = 4.3816 \mu A$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 27^\circ}{11,600}$$

$$= 25.862mV$$

(b) $I_D = 100mA$, $V_D = ?$

$$\Rightarrow V_D = \eta V_T \ln \left(\frac{I_D}{I_o} + 1 \right)$$

$$\Rightarrow V_D = 1 \times 25.862mV \times \ln \left(\frac{100m}{4.3816\mu} + 1 \right)$$

$$V_D = 0.2595 V$$

.Calculate the static and dynamic resistance of a P-N Junction diode, when the applied voltage is 0.25V for Germanium Diode. $I_0 = 1\mu A$ and $T = 300K$.

Given, $I_0 = 1\mu A$, $V_D = 0.25V$, $T = 300K$, $\eta = 1$ for Ge Diode

$$I_D = I_0 \left(e^{V_D / \eta V_T} - 1 \right)$$

$$\Rightarrow I_D = 1\mu A \times \left(e^{0.25 / 1 \times 25.862m} - 1 \right)$$

$$\Rightarrow I_D = \mathbf{15.782 mA}$$

$V_T = \frac{T}{11,600}$
 $= \frac{273^\circ + 27^\circ}{11,600}$
 $= 25.862mV$

For Static Resistance,

$$(a) R_D = \frac{V_D}{I_D} = \frac{0.25}{15.782m} = \mathbf{15.84\Omega}$$

For Dynamic Resistance,

$$(b) r_d = \frac{\eta V_T}{I_D} = \frac{1 \times 25.862m}{15.782m} = \mathbf{1.638\Omega}$$

A silicon diode has a saturation current of 1pA at 20°C. Determine (a) Diode bias voltage when diode current is 3mA. (b) Diode bias current when the temperature changes to 100°C, for the same bias voltage.

Given, $I_o = 1pA$, $I_D = 3mA$, $t = 20^\circ C$, $\eta = 2$ for Si Diode.

$$(a) V_D = ? \quad V_D = \eta V_T \ln \left(\frac{I_D}{I_o} + 1 \right)$$

$$\Rightarrow V_D = 2 \times 25.2586mV \times \ln \left(\frac{3m}{1p} + 1 \right)$$

$$V_D = 1.102 V$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 20^\circ}{11,600}$$

$$= 25.2586mV$$

$$(b) t = 100^\circ C \quad I_D = ?$$

$$I_{02} = I_{01} \cdot 2^{(t_2 - t_1)/10} = 10^{-12} \times 2^{(100 - 20)/10} = 256pA$$

$$\Rightarrow I_D = 256pA \times \left(e^{1.102 / 2 \times 32.155m} - 1 \right)$$

$$\Rightarrow I_D = 7.08 mA$$

$$V_T = \frac{T}{11,600}$$

$$= \frac{273^\circ + 100^\circ}{11,600}$$

$$= 32.155mV$$